



Digital Transformations for
Supporting Next-Generation Labour

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Erasmus+ Programme
of the European Union



CodeSensei

Interactive AI Mentor for a Metaverse Laboratory

Nomination II — Interactive AI application

Team CodeSensei

Igor Sikorsky Kyiv Polytechnic Institute

Team members

Mark Mamenko, Sviatoslav Pavlenko, Denys Iliencko,
Kateryna Shozda, Dmytro Poshytyniuk

nextstudy.eu

Mark Mamenko

Igor Sikorsky KPI

Role in the project

Backend / Team Lead. Clean Architecture, memory-aware .NET proxy, Gemini LLM integration.

Sviatoslav Pavlenko

Igor Sikorsky KPI

Role in the project

UdonSharp Developer. Interactive terminal UI and GET-only VRChat networking.

Denys Iliencko

Igor Sikorsky KPI

Role in the project

AI / Prompt Engineer. System prompts and OOP context so answers stay useful in VR.

Kateryna Shozda

Igor Sikorsky KPI

Role in the project

QA / C# Tester. NUnit suite (AAA), parsing logic and backend stability.

Dmytro Poshytyniuk

Igor Sikorsky KPI

Role in the project

Integration / VR Tester. Prefab assembly in Unity and HTTP checks inside VRChat.

How we work

Agile team of five. Unity client and .NET API in parallel, synced over Git. Cross-platform stack (Rider, macOS).

THE PROBLEM

No mentor in the VR lab

Department MetaLab on VRChat · NEXT-Study Metaverse

CodeSensei is an NPC-style AI assistant at a virtual terminal. Students get OOP explanations, help writing code, and instant feedback without leaving VR — a stand-in for a consulting teacher during remote practice.

PROJECT GOALS

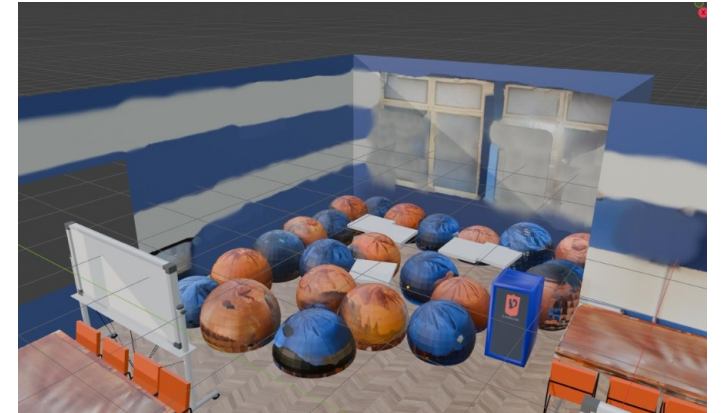
- UdonSharp (C#) client inside VRChat
- Secure .NET proxy to the LLM API
- Presence of a teacher in MetaLab
- Clean Architecture + DDD for later scale
- Unity package any NEXT world can import

Selected Location

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MacPaw AI Lab / MetaLab classroom

The application lives in the department's virtual laboratory on VRChat: seminar rows, critique boards and an informal lounge. The terminal sits at a workstation so the mentor is present in the same room as the class.



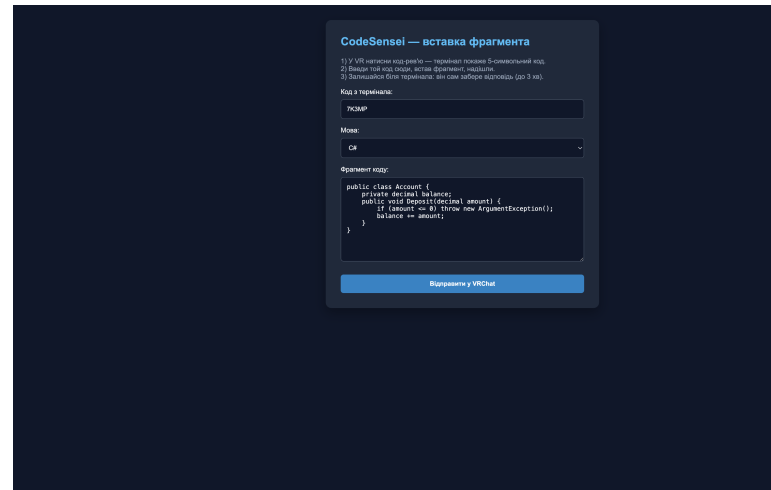


UdonSharp terminal



Unity prefab: UI terminal, presets and GET-only VRCStringDownloader for VRChat SDK3.

Paste + ticket flow



Five-character ticket, public /paste form, inbox the prefab already polls.

Clean Architecture API



Protected .NET proxy to Gemini. Keys never enter the game client. NUnit on the domain.

Tools and software

.NET (C#) Clean Architecture proxy · UdonSharp / VRChat SDK3 · Unity prefab · Google Gemini (LLM analog of OpenAI API) · System.Text.Json · NUnit (AAA) · Docker / Render · Git.

Development process

1) Split Unity client and API. 2) Lock a GET-only VRChat contract. 3) Isolate the LLM behind prompts and a proxy. 4) Cover the backend with NUnit. 5) Ship CodeSensei.unitypackage for MetaLab.

VRChat integration

HTTPS host codesensei-d5zi.onrender.com. VRCStringDownloader only (Udon cannot POST). Allow Untrusted URLs. Polling ≥ 5.5 s. Ticket + /paste + inbox for reviews.

Technical challenges

POST is impossible from Udon, so reviews use a 5-character ticket. Pending tickets stay out of the inbox. Render Free is woken via /health. Answers wrap to 55 characters for the VR screen.

Educational use

In NEXT-Study Metaverse a student stands at the MetaLab terminal, opens an OOP topic or submits a snippet, and receives mentor feedback as if a consulting teacher were in the room. The same text is synced for the group.

User experience

Walk the laboratory, press a preset, or generate a ticket, paste code on the web form, and read a formatted reply on the shared VR terminal. Keys never leave the server.

Further development

Drop the prefab into other NEXT worlds, persist tickets, swap the LLM provider (architecture already isolates it), add languages, and keep the NUnit safety net.

Demo and Project Materials

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Project demo

<https://codesensei-d5zi.onrender.com/paste>
<https://codesensei-d5zi.onrender.com/health>

Project materials

GitHub: github.com/mamenkomarkus-creator/CodeSensei-NEXT
Prefab: [client/CodeSensei.unitypackage](#)
Docs: [docs/SUBMISSION.md](#) · [docs/api.md](#) · [docs/NEXT-INTEGRATION.md](#)

Additional resources

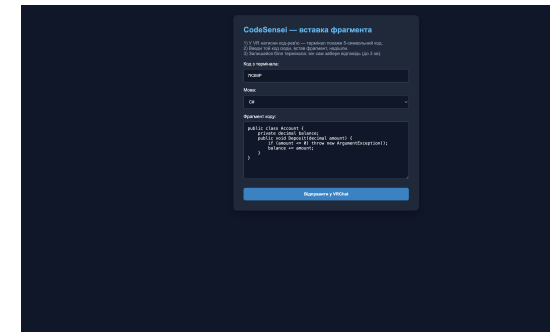
VRChat: hello.vrchat.com NEXT: nextstudy.eu
Live API: codesensei-d5zi.onrender.com/paste



GitHub



Demo



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Thank you!
Questions?

